

CUSTOMER x
Hamarashops.ai:

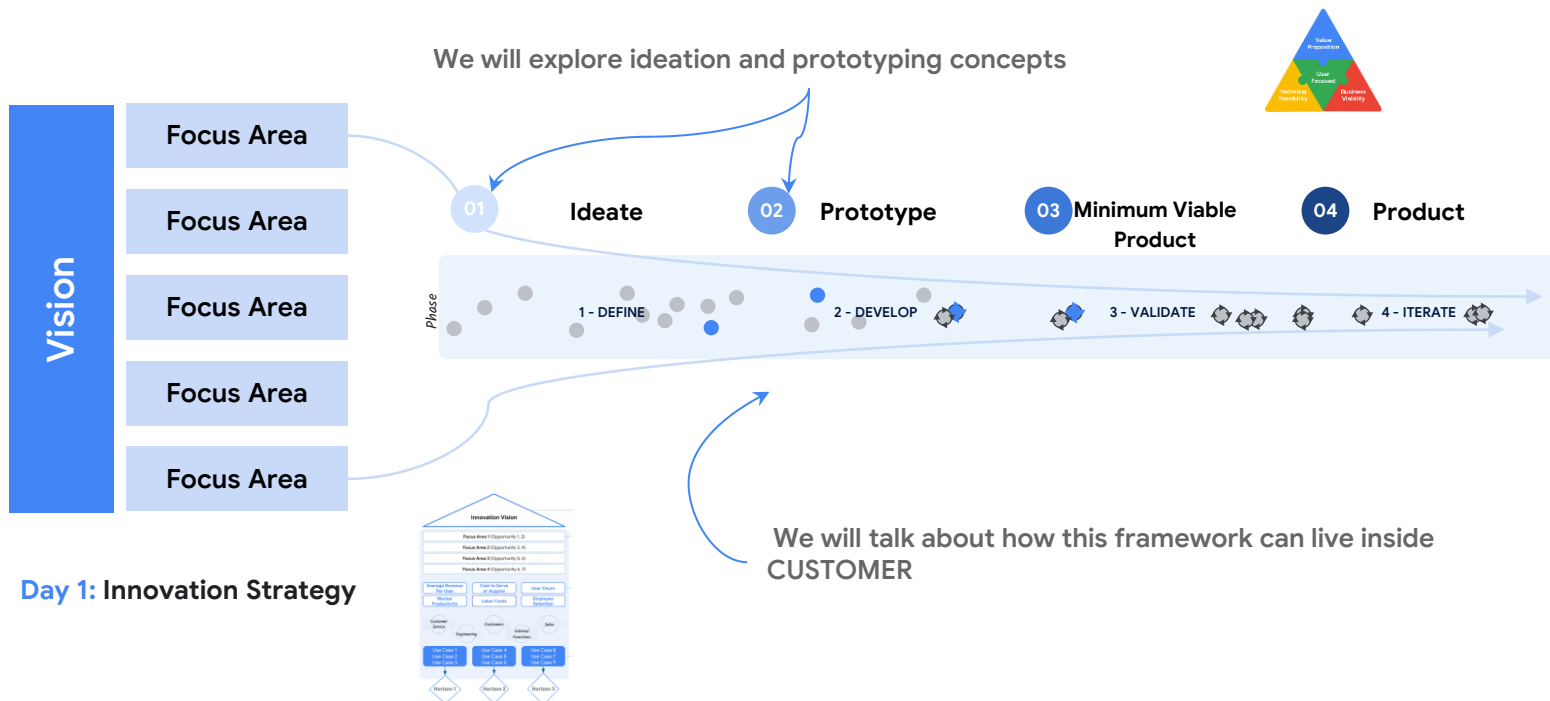
AI Innovation Workshop

Interactive session to rethink
healthcare with Hamarashops.ai.

Google Cloud
Consulting



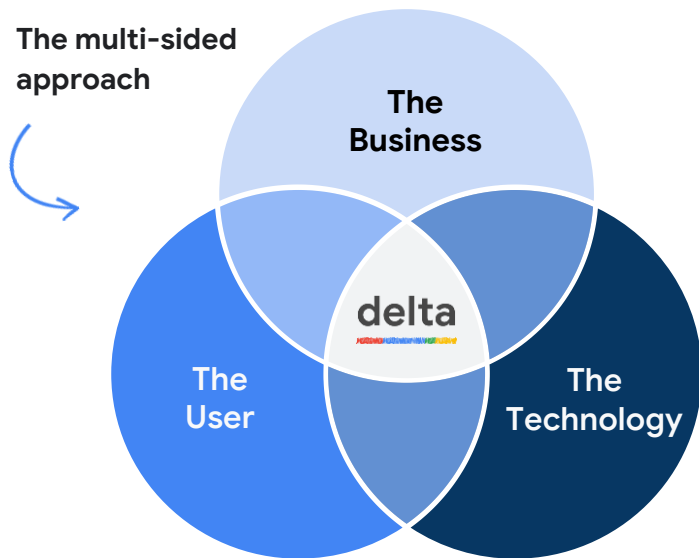
Hamarashops.ai Goals: Explore these questions by putting the framework into practice



Day 1: Innovation Strategy

Hamarashops.ai Delta team:

Innovation and value driving use case development is successful only when we consider the business, the user and the technology.



Solving for the Business

- Top revenue growth (i.e., ARPU)
- Lower acquisition and service costs
- Reduced churn and improved upsell

Solving for the User

- User focused and intuitive
- Brand creation and loyalty
- Increased satisfaction

Solving for the Technology

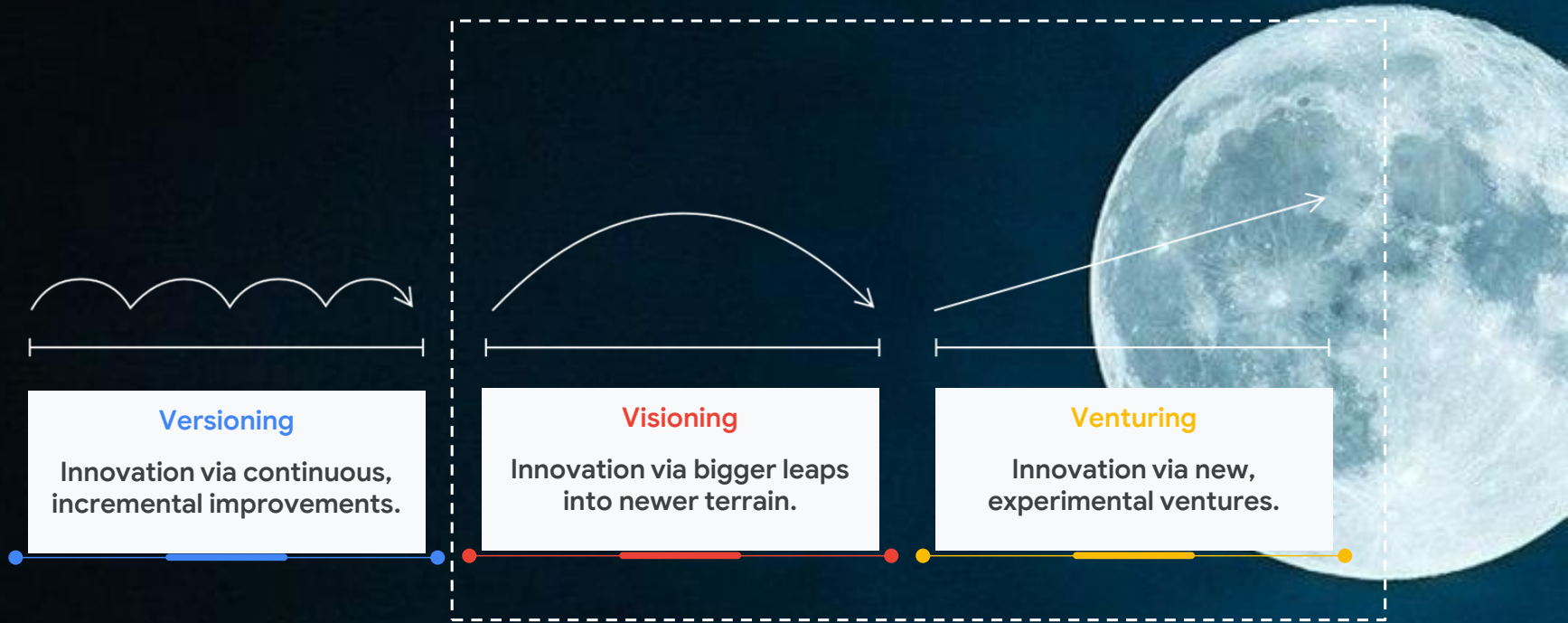
- Data driven
- AI powered
- Scalable and efficient

Why it's right

During planning and execution, the lines between the business, user, and technology often get blurred and overlooked.

Ensuring that business, user, technology needs are considered at the onset ensure that the best capabilities are integrated into a unified approach and blended team.

Visioning at Hamarashops.ai:



Exciting times: Hamarashops.ai



Artificial Intelligence in Healthcare



What do these two have in common?



Software Engineers

**Engineers and
doctors spend more
than 30% of their
time with
documentation.**



Doctors

Similarly, there is proven potential for AI use cases in healthcare

Source: <https://doi.org/10.2196/59617>,
<https://doi.org/10.1126/science.adk6139>



Analyze images

Use of AI models to analyze image data for decision support for clinicians (e.g., radiology, optometry)



Create reports

Automatically create documentation (e.g., doctor's letters/discharge summaries, surgical reports, medical coding)



Structure data

Processing of unstructured data (e.g., doctor's letters/discharge summaries, guidelines, studies) to generate recommendations to support medical experts

Example 1:

ARDA¹ helps healthcare workers detect diabetic retinopathy



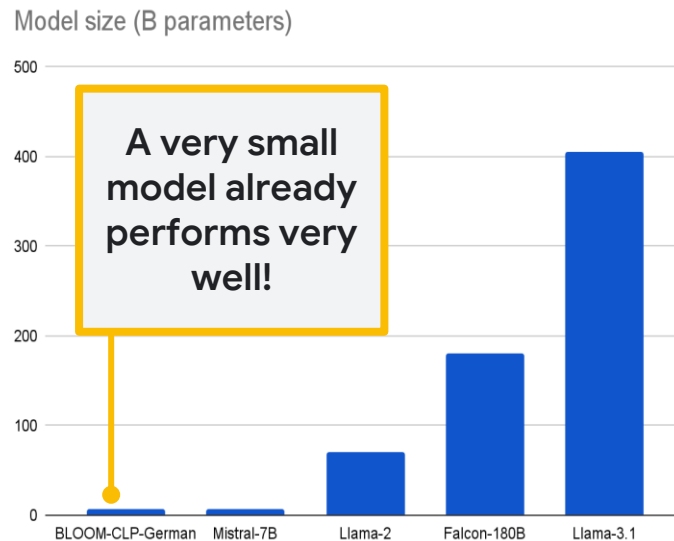
- Ophthalmologists labeled more than 100,000 de-identified retinal scans.
- A ML-model is now able to detect diabetic retinopathy in retina scans.
- This has been used to conduct more than 200,000 retinal screenings in India, EU, Thailand and US.



Example 2: Using AI to write doctor's letters



- Researchers trained multiple custom LLMs to assist medical professionals with writing using local infrastructure in a secure hospital environment.
- Human experts confirmed that 93.1% of the letters generated by a custom model based on BLOOM-CLP-German were usable as-is or with minor alterations.



Example 3:

Generation of therapy recommendations for molecular tumor boards



Structure data

- Researchers configured an LLM (Google Gemini) to generate therapy recommendations for molecular tumor boards (MTB).
- By processing a large amount of publicly available data (especially Pubmed) in a manner analogous to the approach of precision oncologists, the model achieved 94.7% agreement with MTB experts.

Literature
Screening

600+
Documents

Literature
Review

20-30
Documents

Therapy
Options

5-10
Options

MTB
Recommendation

2-3

What healthcare providers can learn from Google



Create human-centric design



Find a real problem to solve



Build for explainability

Respect the user



Human-centric
design



Hello, my name is

Julia

And I am a

Clinician

My skills and experience

- High comfort with molecular pathology data and translation into therapy concepts.
- Good basic understanding of technology and medical software.
- Open to adopting new technologies that improve patient care and efficiency.

My objective is

- Provide the best possible treatment for her patients based on their individual profiles.
- Stay up-to-date with the latest research and clinical trials in precision oncology.
- Improve patient outcomes and quality of life.

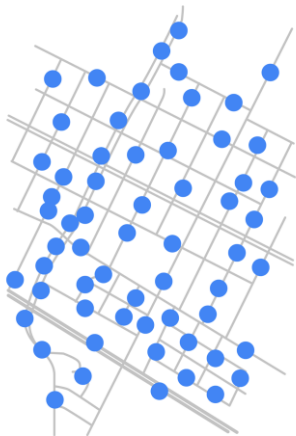
My biggest pain points are

- Limited time to thoroughly analyze and explore all potential treatment options.
- Availability of data in structured formats that I can easily analyze.
- High administrative burden and manual documentation instead of patient care.

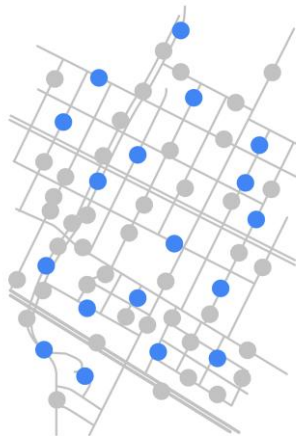
Don't address every pothole in the road—focus on one (critical) journey



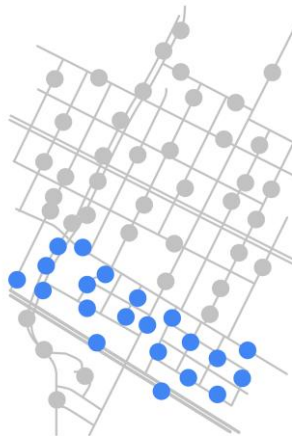
Find a real
problem



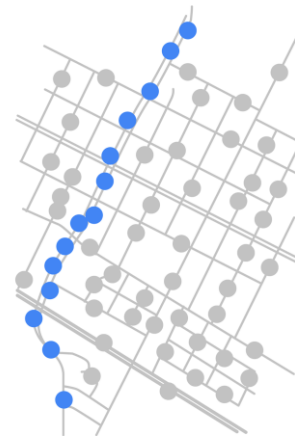
Potholes in your
roads



Top reported
issues



Feature set



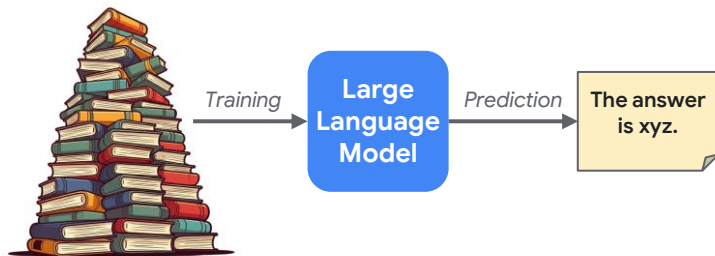
One journey ✓

Explain outcomes of your AI system to build user trust



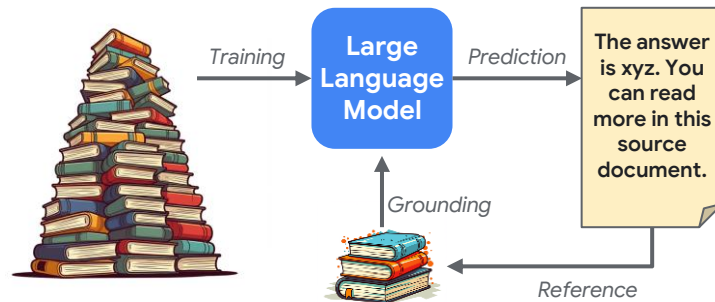
Build for
explainability

By default, AI systems cannot explain their reasoning



- AI models are trained on a large amount of labeled data
- They use mathematical models to predict the most likely answer ("label") to a question ("feature")
- By default, these predictions are not explainable to users

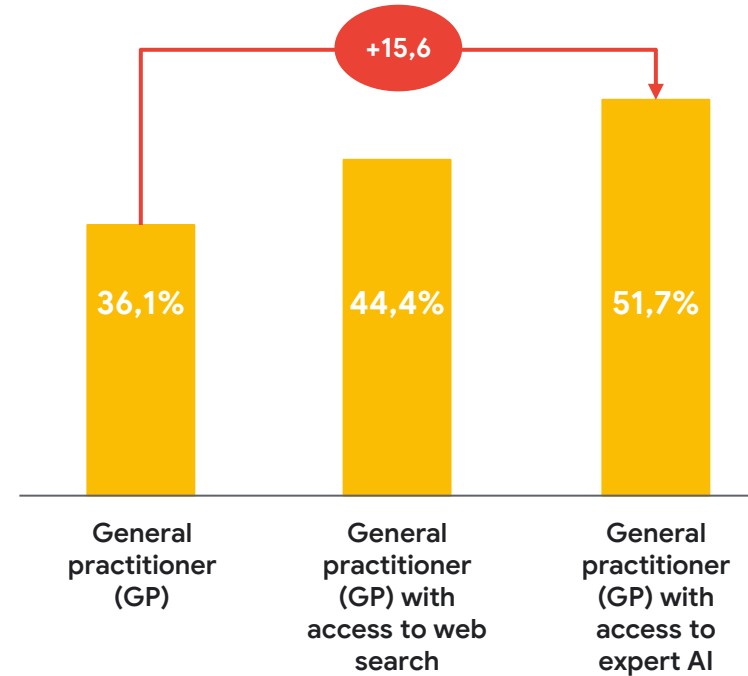
By grounding in data, we validate output and can explain reasoning to users



- We add grounding data (e.g., a medical guideline), and instruct the model to always use this data for predictions
- The model can cite and reference its sources, lending credibility with users and reducing "hallucinations"

AI is not a
substitute for
specialist
medical care,
but rather a tool
to improve it

Percentage of correct differential diagnoses
on 302 real-world cases from NEJM,
randomized and expert-reviewed.



Example focus areas

Customer Service: Focusing on providing responsive, empathetic, and efficient support to patients and their families.

Patient Experience & Quality: Ensuring safe, effective, and person-centered care that consistently meets or exceeds patient expectations.

Operational Excellence: Optimizing processes, resources, and workflows to deliver high-quality care efficiently and sustainably.

Treatment Innovations: Advancing medical care through the adoption and development of new therapies, procedures, and medical technologies.

Workforce Development: Investing in the growth, well-being, and empowerment of staff

AI-first Hospital: Leverage cutting-edge technologies, data analytics and AI to enhance patient care, operations, and innovation

Health Equity and Access: Reduce health disparities to ensure all members of the community have fair and just opportunities

Data monetization, data security, automation of clinical coding, student experience

Today

O

Objective

Aspirational goal
part of the org's larger one

set for longer time period
(e.g., one or multiple years)

have no more than 3-5

KR

Key Result

Clear milestone
towards an objective

set for specific time
frame
(e.g., quarter or year)

have no more than 3-5
for each objective

Day 2

A

Activity

Project or task
towards a key result

might change frequently
(e.g., within quarter)

has clear owner(s)

O

Objective

Why

are we doing
this again?

KR

Key Result

What

is the dent we're
looking to make?
(impact!)

A

Activity

How

are we going to
achieve that
result?

“ We're going to **A** launch feature x in
order to get **K** 95% of P2P payments done in <5s
because we want
to **O** Be the fastest payment app in India ”



Objective

Be the fastest payment app in India

KR

Key Result

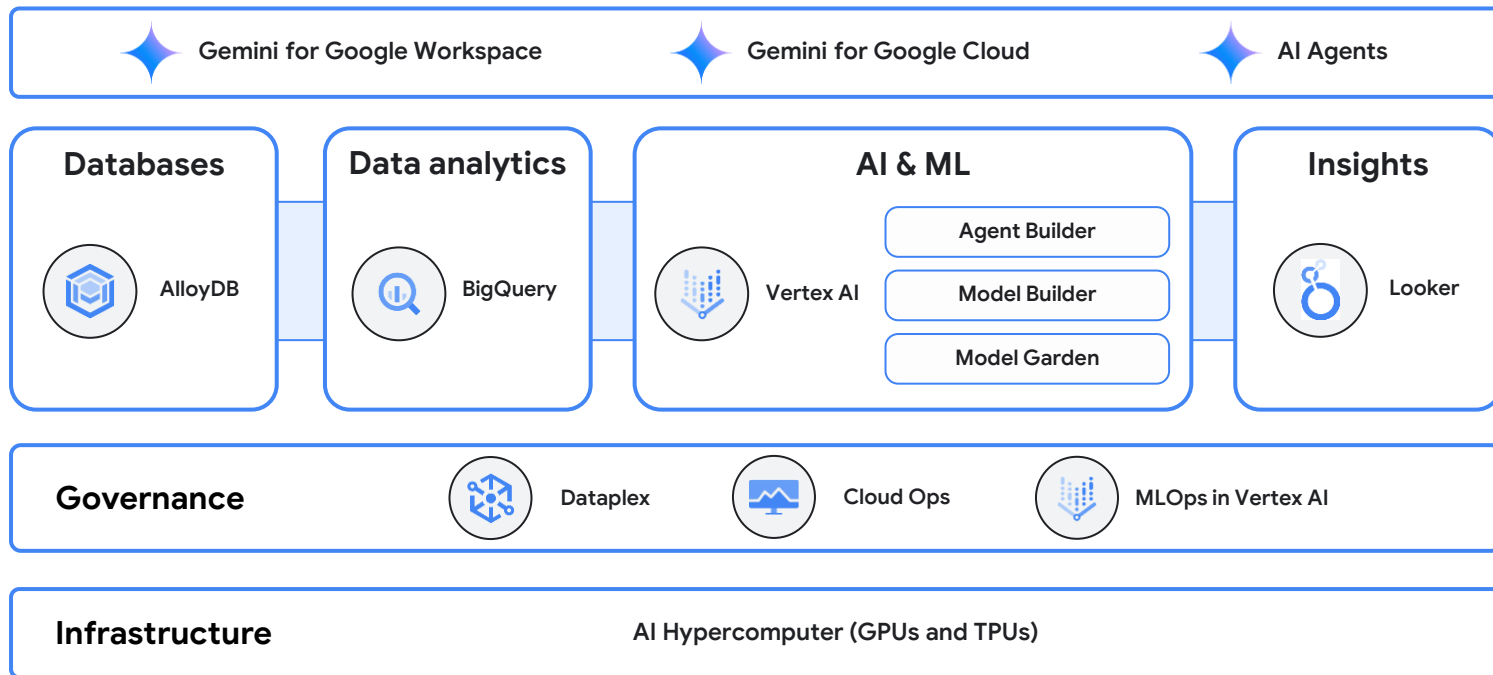
95% of P2P payments done in <5s

A

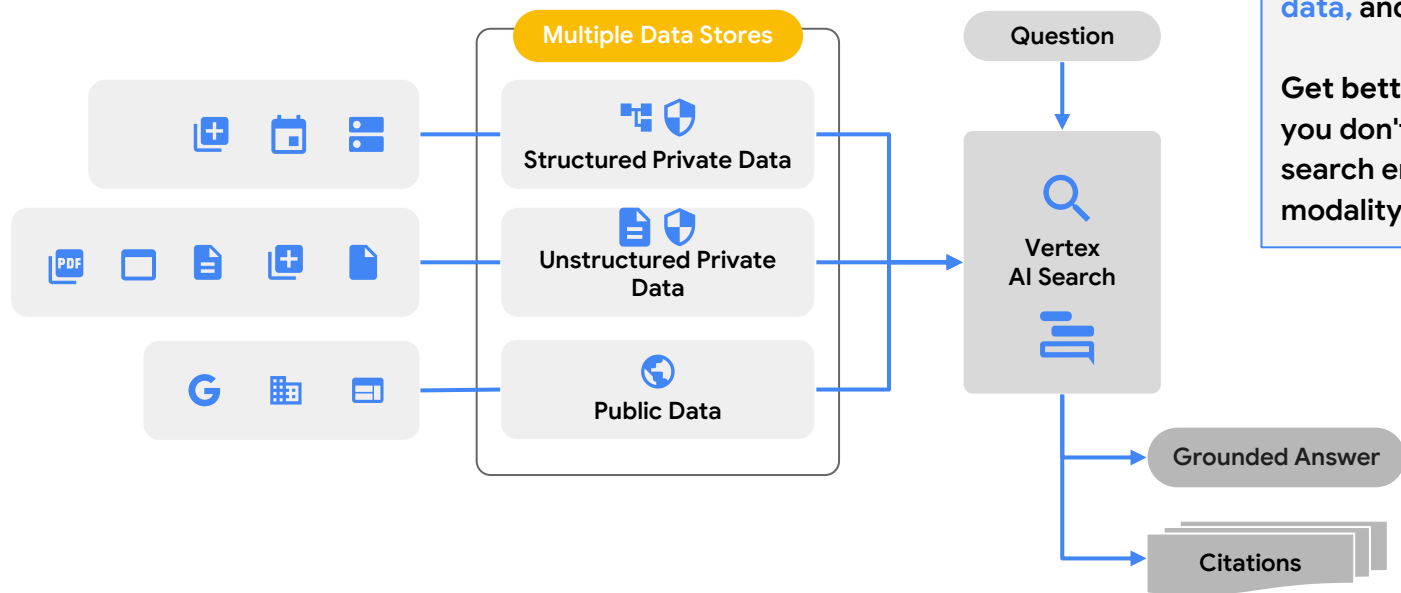
Activity

Launch project Neverslow, phase I

A unified platform from data to deployment and for all your predictive, generative, and agentic needs



With Vertex AI Search, we provide a unified experience across structured, unstructured, and public data

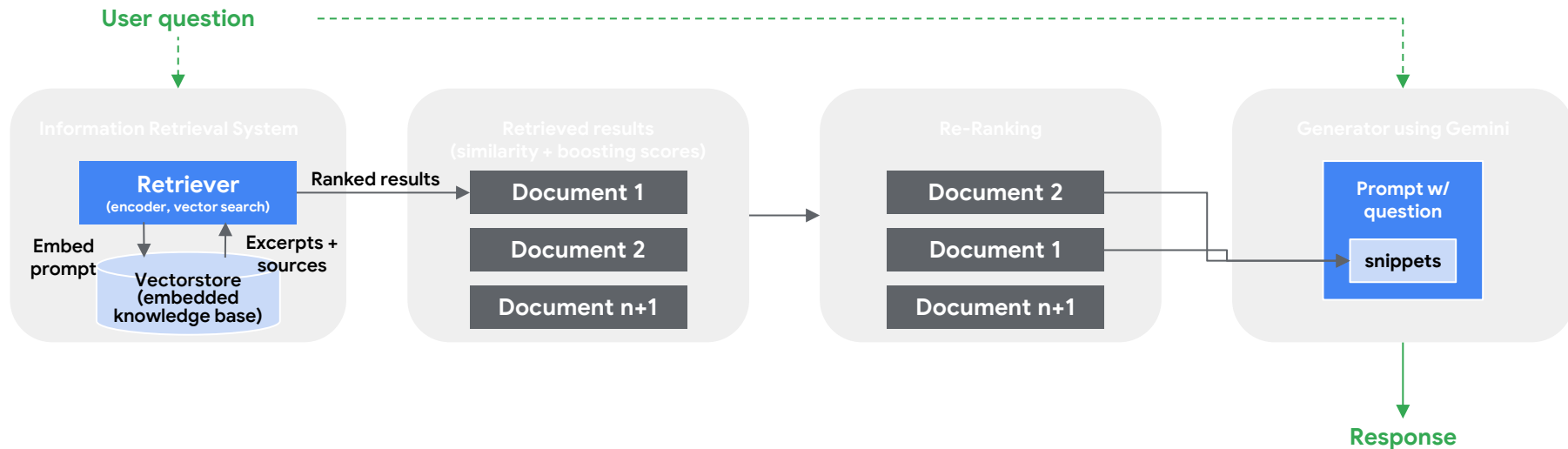


Run a single query across
structured data, **unstructured data**, and **public web pages**

Get better quality search results -
you don't have to build separate
search engines for each type and
modality of data and then integrate

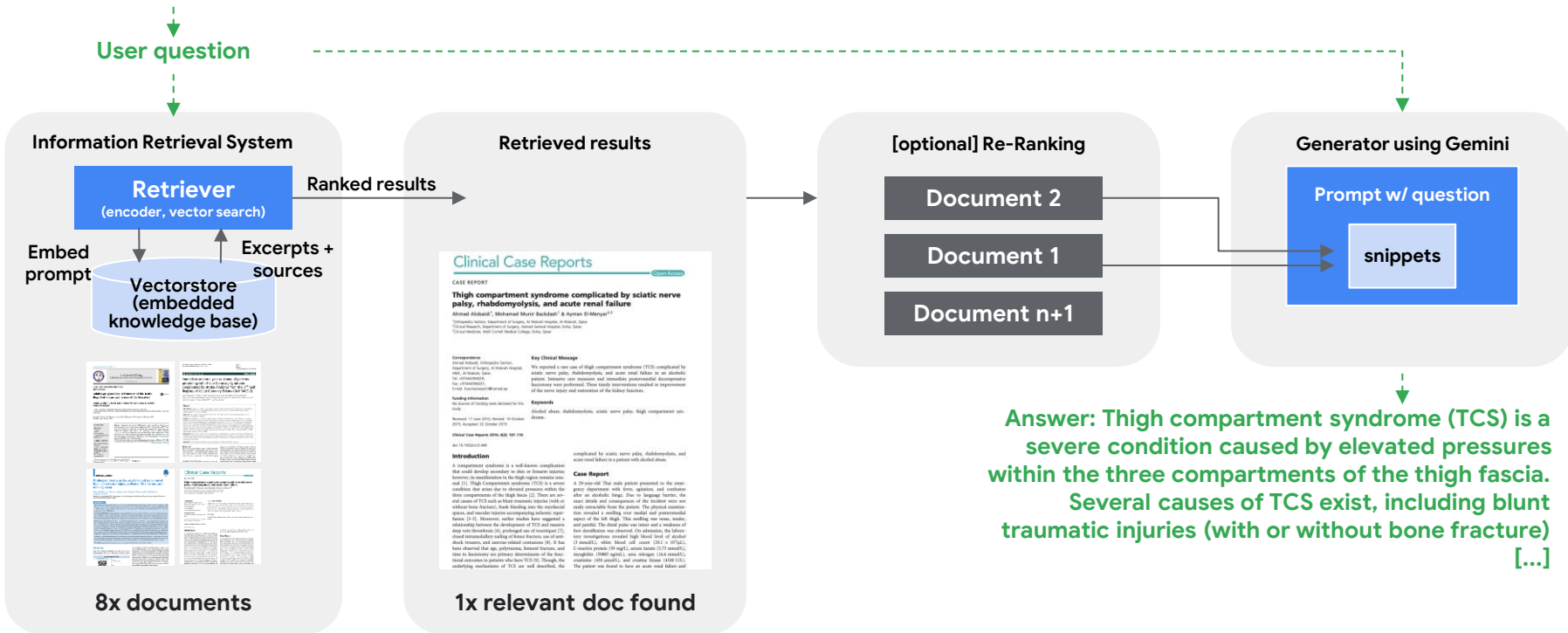
Deep dive into RAG systems

RAG applications



Vertex AI Search in healthcare –Hamarashops.ai architecture

What are the causes of thigh compartment syndrome (TCS)?



Unlock enterprise expertise for clinicians, nurses, claims, revenue cycle members, and employees

Google Gemini Enterprise empowers healthcare workforce with

Unified Google-quality search experience across all healthcare operational, financial, and clinical data

Google's latest AI innovations such as multimodal, NotebookLM, Imagen, and more

Centralized destination for employee agents across OOTB, 1P, and 3P agents

Enterprise grade security, privacy, and compliance

MedLM

MedLM Demo

Choose a topic to explore

Rapid heartbeat

Cause of constipation

Loss of balance

Pancreatitis

Endocarditis

Baby reflux

Molar pregnancy

Baby reflux

Seasonal affective disorder

Irregular periods

Mumps

Color blindness

Breast cancer

Kidney stones

 What are the first signs of passing a kidney stone?



The first signs of passing a kidney stone are typically sudden, severe pain in the lower back or side, which may radiate towards the groin or genitals. Other symptoms may include: nausea, vomiting, and blood in the urine. The pain from a kidney stone is often compared to the pain of childbirth.

Review Evaluation Results ▲

Reflects clinical and scientific consensus ✓

Not harmful ✓

Low likelihood of harm ✓

Correct reading comprehension ✓

Correct recall of knowledge ✓

Correct rationale ✓

Relevant information ✓

Comprehensive information ✓

No demographic biases ✓

Choose a topic from the left

NotebookLM Enterprise

NotebookLM Enterprise is your AI research and writing tool, designed to help you find, summarize, and gain insights faster while grounded in your organization's shared knowledge base.

All using  Gemini

Sources:

Websites

Slides

Docs

PDFs

Text

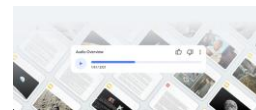
YouTube URL

Audio

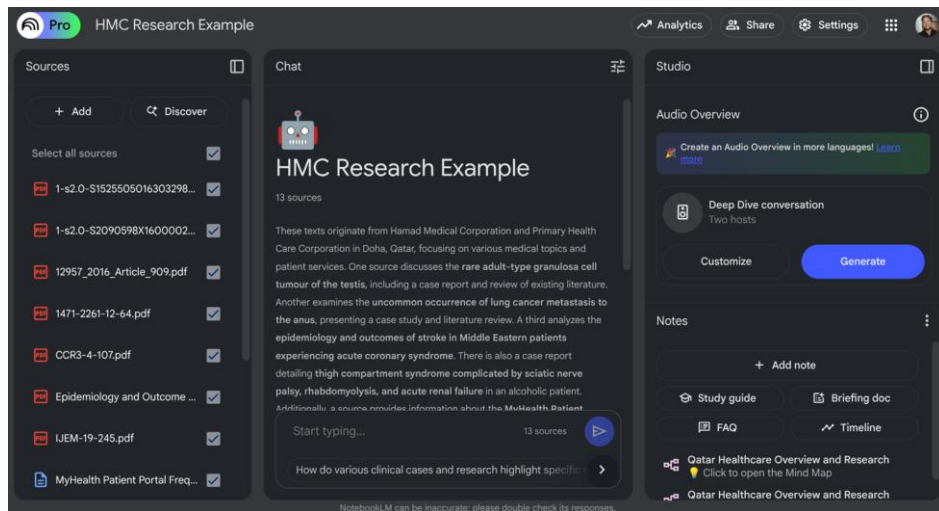
Microsoft Docx + PPTx



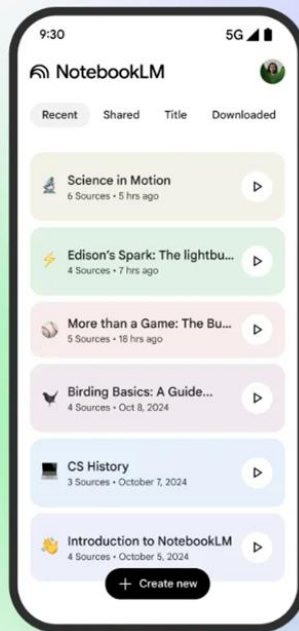
NotebookLM can help you get a deeper understanding of whatever you're working on.



Audio Overviews turn your documents into engaging audio discussions.



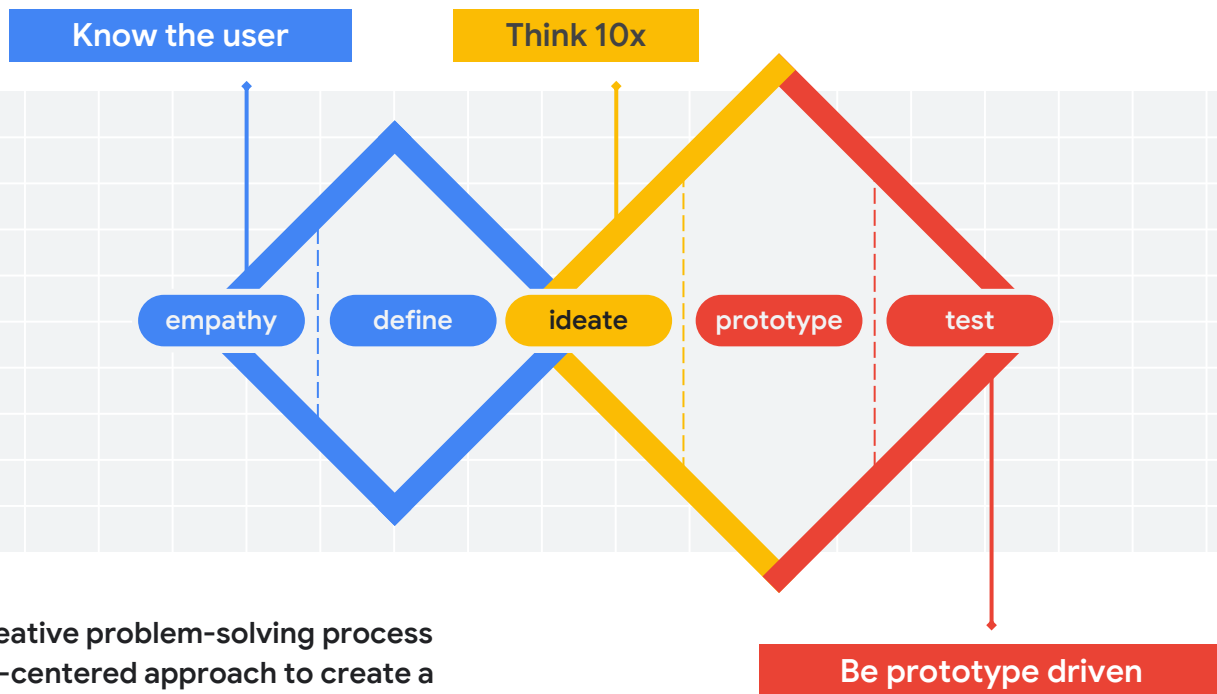
NotebookLM App



The app is now available on the [Apple App Store](#) (iOS 17+, iPhone and iPad) and [Google Play Store](#) (Android 10+, phones and tablets).



Hamarashops.ai design thinking framework **idea lifecycle**



Design thinking is a creative problem-solving process that focuses on a user-centered approach to create a solution that is technologically and economically feasible.

Stages of the patient journey:



Before visit

Provide personalized,
convenient, preventative
pre-care service



During visit

Improve access and
transparency and
transform **care delivery**



After visit

Empower patients with **post-care** tools and resources to
improve outcomes



Connected end-to-end patient journey //

Exchange of information



Exercise: Use case ideation **brainstorming**

Step 1

Individual brainstorm

Write down your ideas on post-its and put them on the board

Step 2

Group Discussion + Yes And

Share your ideas and increase others' ideas with your "Yes and..."

Step 3

Clustering

Step 4

Voting

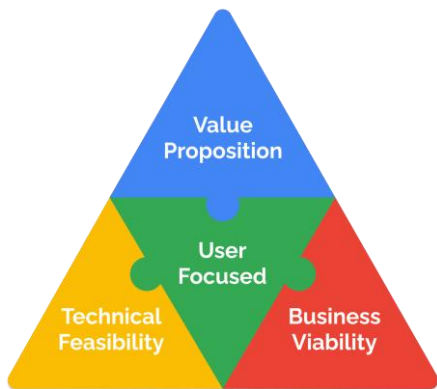
Vote individually for the top 5 augmented ideas you like using icons



The use case canvas

Use cases should capture input and drive alignment on the most critical questions influencing the ability to solve user needs and business goals.

We use the use case canvas to explore all the aspects of innovation with the user at the center across our end-to-end approach.



User Focused

Who are our users and most valuable customers?

Who is the most important user target?

What pain points and needs should we solve for the user?

What would delight the user?

Value Proposition

What 'need' exists that this use case can provide a unique, differentiated solution?

What is our competitive advantage, differentiation or right-to-win?

What value is created?

What are the internal and external threats? Opportunities?

Technical Feasibility

What considerations need to be made regarding current and future technology?

What are the different types of data?

What are the data requirements? Is it available, complete, and accurate?

How might we derive actionable insights from data?

What leverageable capabilities do we already have or are under development?

Business Viability

How will this solution impact our business? What does the data show us?

What are the people and process requirements to support the initiative?

What commercial impact and business value will we drive?

What are the investment requirements, and how could we execute?

Post-workshop: Developing a prioritized catalog of use cases is key

Use Case Details					Evaluation Criteria			
Use Case Details	Exec. Sponsor	Functional Org.	Priority / Timing	Archetype / LLM	User Impact & Focus	Value Proposition & Market Potential	Technical Readiness & Requirements	Business Viability & Requirements
Customer Service Infobots	-	-	Q1 '24	-	L / M / H	L / M / H	L / M / H	L / M / H
Knowledge Worker Enterprise Search	-	-	Q1 '24	-	L / M / H	L / M / H	L / M / H	L / M / H
Contract document ingestion & creation	-	-	Q2 '24	-	L / M / H	L / M / H	L / M / H	L / M / H
Code accelerations & assistant for engineers	-	-	Q2 '24	-	L / M / H	L / M / H	L / M / H	L / M / H
Accounts payable	-	-	Q3 '24	-	L / M / H	L / M / H	L / M / H	L / M / H
Marketing content creation	-	-	Q3 '24	-	L / M / H	L / M / H	L / M / H	L / M / H
Employee Infobots	-	-	Q4 '24	-	L / M / H	L / M / H	L / M / H	L / M / H
...other use cases	-	-	Q4 '25	-	L / M / H	L / M / H	L / M / H	L / M / H

We do search but also lot of research!

Alphabet



Google Ventures
Venture & Capital
Funding



Loon
Flight
Connection



Google X
Innovation Lab
& Research



Search
Advertising
SEM



Google Cloud
Healthcare and
Life Sciences



Waymo
Self Driving
Vehicles



Fiber
High Speed
Internet Services



Calico
Longevity
Research



YouTube
Internet
Video
Service

Google Health

Google Health
AI and Consumer
Health Products



SideWalk Labs
Solving
Big Urban
Problems



Wing
Drone-based
Delivery of Freight



Verily
Improving
Quality of Life



Maps
Mapping,
Location Services
& Logistics



Google Marketing
Platform
Data Analytics
Suite of Tools



Jigsaw
Global Online
Security Solutions



CapitalG
Private
Equity



DeepMind
AI &
Machine Learning



Devices and
Services
Pixel, Nest
Chromecast



Android
Mobile Operating
System



hamarashops.ai

AI POWERING YOUR BUSINESS

Mr. DHEERENDAR

CEO



Hamarashops.ai is an application player in AI race



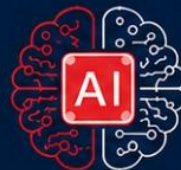
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Website
<https://www.hamarashops.ai>



**SMART SOLUTIONS.
POWERED BY AI.
BUILT FOR YOU.**

Thank you

